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Autonomous navigation of non-perception firefighting robot through CCTV-informed vision sharing

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ABSTRACT

Deploying smart firefighting robots indoors is a prospective measure to realize fast-response and zero-casualty firefighting operations. Most existing firefighting robots require manual control or pre-input information for navigation. This work proposes a CCTV-informed autonomous navigation system for robot to reach target flame via the safest and fastest route. This proposed system uses the vision of building CCTV-camera network and provides a full-process navigation pipeline to guide imperceptive firefighting robots. Demonstrations show that the navigation system enables the robot to have a complete perception of floorplan and understand the evolution of fire scenes. The robot can autonomously move towards the target flame even if carry-on vision sensors are blocked or damaged. Finally, the dynamic system responsiveness is validated to prove its effectiveness in addressing randomly moving obstacles. This framework can be further integrated with smart building maintenance system, which offers low-cost, early-response, and more resilient solutions for firefighting and safety patrol robots.

1. Introduction

Given rapid urbanization and globalization, the increasing complexity and integration of modern building designs make it difficult for firefighters to access fire sites and implement targeted actions (Khan et al., 2022). To mitigate the risks associated with direct human involvement in hazardous fire environments, many emergency departments and institutes are turning to the development of firefighting robots to perform dangerous firefighting tasks (Sawant et al., 2020). The use of firefighting robots offers a promising solution for achieving fast-response and zero-casualty firefighting operations (Lattimer et al., 2023). However, an autonomous firefighting pipeline, including fire detection, fire source localization, environmental perception, path planning, path following, and fire extinguishing, remains an extremely challenging task to be realized on firefighting robots (Chaoxia et al., 2023).

Over the years, numerous scholars and industries have been working on the intellectualization and automation of firefighting robots. The development of firefighting robots has evolved through three stages: manual control level, partial autonomy level, and full intelligence level. Most applied firefighting robots are stuck in the manual remote control, where operators are required to direct every movement of the robot and its loads (Liu et al., 2016; Zhang et al., 2023). Although these robots are typically equipped with cameras and temperature detectors, they lack advanced environmental perception capabilities. The data collected from these sensors are primarily used to avoid obstacles, rather than for comprehensive situational awareness (Wang et al., 2024). Recent advancements in artificial intelligence (AI) have enabled firefighting robots to achieve a certain degree of autonomy. For instance, the occurrence of fire patrol robots can effectively reduce the response time of firefighting actions by providing 24-hour monitoring and operating along predefined routes (Chen et al., 2023). These robots transmit video or temperature data back to headquarters and can activate firefighting systems once a hazard or flame is detected (Zhang et al., 2022a; Zhang et al., 2022b). However, these partially autonomous robots still rely on initial human commands and predefined areas, as environmental maps and states are generally provided to them for specific tasks. While this reduces the need for real-time environmental perception, it limits the

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ability of these robots to adapt to dynamic changes in the environment. The development and application of firefighting robots at the full intelligent level is still a gap. There is an urgent need to transition from teleoperation or partial autonomy to full autonomy (Spurny et al., 2021).

The fire scene is one of the most challenging environments for robots, because there is very limited information from fire scene; the smoke can block almost all visual sensors (including LiDAR); and hot flame and gas can damage sensors in building and robot. For an automatic firefighting mission, key information such as the location of fire sources and the distribution of obstacles is critical. As a result, environmental perception is a fundamental task for robotic navigation in fire environments. Multiple detectors, such as vision cameras (Yang et al., 2017), stereo cameras (Zhang et al., 2021), thermal cameras (Zhu et al., 2020), LiDAR (Guo et al., 2009) and event cameras (Ding et al., 2025), are normally mounted on the robot to help it understand environmental information. These perceptual sensors access vital data on distance, temperature and other environmental factors, which allows firefighting robots to move along an optimized route with a detailed environmental map. However, a single sensor has the limited ability to completely reproduce complicated fire scenes. An ancillary strategy decoupled from robotic motion is needed to aid vulnerable instruments in recognizing fire environment and to avoid failure or damage of energy-intensive, sophisticated instrumentation.

Multi-sensor data fusion is proposed to leverage the complementary strengths of multi-modal data (Luo et al., 2009). The integration of cameras (both vision cameras and thermal cameras) and LiDAR is the most common approach for acquiring comprehensive environmental data (Kim et al., 2015). However, when robots are approaching flames, these laser and vision sensors may encounter degradation or malfunction under high-temperature and smoky fire environments (Phillips et al., 2017), resulting in the disorientation and even paralysis of robots. Previous literature indicates that the signals from LiDAR start to be attenuated within visibility of 4 m whilst visual and thermal cameras are interfered (Starr & Lattimer, 2014). This adversely affects the robotic dependability of close-range firefighting. It entails early-stage fire forewarning and intervention by robots. The role of firefighting robots is not limited to machine firefighters, but can also serve as autonomous responders to fires before they develop dramatically. Moreover, obstacle mapping using LiDAR and stereo cameras relies on the Simultaneous Localization and Mapping (SLAM) algorithm, which means a prior reconnaissance of fire fields is necessary (Hong, 2022). As its name implies, the SLAM method maps the unknown environment while exploring it. Hence, firefighting robots can only be deployed in fire environments once unambiguous maps of fire sources and obstacles are available.

Today, CCTV cameras appear in each corner of the buildings serving as a universal and essential infrastructure. This camera network provides real-time visual data covering large areas. As huge advances have been made in computer vision (CV) and pattern recognition, CCTV cameras are increasingly used to assist with architectural management (Hu et al., 2023; Pardamean et al., 2019), fire risk warning (Pincott et al., 2022; Zhang et al., 2022a), disaster management (Jeon et al., 2024; Muhammad et al., 2018), tunnel fire evaluation (Zhang et al., 2024) and pedestrian evacuation (Ding et al., 2023). A major issue for autonomous navigation is to make robots determine their location, identify the destination, and plan an optimal path (Alatise & Hancke, 2020). CCTV cameras offer a strategic advantage over the onboard vision sensors of firefighting robots, providing broader coverage and better perspective. By sharing visual data between robots and CCTV cameras, real-time fire risk assessments and detailed environmental information can be obtained, helping robots avoid areas with unknown hazards in advance (Kim et al., 2024). This integration allows an imperceptive robot to implement firefighting actions with a clear understanding of obstacle distribution and target locations.

This paper introduces an innovative navigation architecture for firefighting robots by sharing visual information and situational data from the CCTV camera, which intelligently integrates fire detection, autonomous robot guidance and visual servo control. The major focus of this work is to achieve rapid response and containment of early-stage fires, where the autonomous robotic system operates to enable proactive assessment and containment. It contributes to the firefighting field by making the following progress:

1) This proposed methodology allows completely blind robots to possess the ability of independent motion and automatic navigation in unexplored fire fields. It generates comprehensive and real-time updating environmental information without aforehand search under unknown situations.

2) With vision sharing and data interpretation from the CCTV camera on a higher and completer perspective, firefighting robots can accurately localize fire sources and calculate optimum routes, even when flames are obstructed by obstacles.

3) This system also facilitates quick implementation of fire suppression at the initial stage of fire development, as early fires can be promptly identified through CCTV footage. Instead of replacing mature navigation methods, our navigation method focuses on a supplementary framework that carries out primary environmental perception and map construction.

4) By utilizing the widespread availability of CCTV cameras, the method allows for seamless and cost-effective integration of firefighting robots into urban and industrial environments, significantly enhancing emergency response and firefighting efficiency.

The rest of the paper is organized as follows. Section 2 reviews the relevant studies about autonomous robot navigation and firefighting robots. Section 3 details the overall system framework and its submodules, including a discussion of the structure and function of each module. The small-scale test, real-scale practical validation and dynamic responsiveness validation are carried out in Section 4, and Section 5 concludes the efforts.

2. Literature review

2.1. Environmental perception of robots

Environment perception is a basic and critical technology for mobile robots, where fire detection is the primary and urgent mission for firefighting robots. Onboard sensors play a crucial role in enabling robots to understand their surroundings, assess their state and identify mission targets (Meng et al., 2016). Various types of sensors, including but not limited to vision, thermal infrared, humidity, and gas sensors, have been widely adopted to detect and position fires in previous literature (Jain & Srivastava, 2022). In general, traditional heat-based and smoke-based fire detection sensors operate when the temperature value or smoke density is above a fixed threshold, which increases the possibility of misinformation and non-response. Additionally, the performance of the sensors highly depends on proximity to fire sources and needs to be placed near fire sources for effective detection (Zhao et al., 2024). The inability of fire localization is as well a major drawback unless costly high-density sensor networks are constructed.

CCTV cameras with their widespread installation in buildings and industries provide a convenient approach to directly access real-time images of high-risk areas. Therefore, vision-based fire detection has become a prominent research topic, particularly using machine learning methods. Flame features, including color features, edge features and texture features, are the key elements to be identified and extracted for a vision-based fire detection algorithm (Emmy Prema et al., 2018). Though numerous scholars have contributed to the identification and description of fires using machine learning methods, they are still constrained by simple flame features and challenged by dynamic flame behaviors (Li & Zhao, 2020; Sharma et al., 2020). The emergence of convolutional neural networks (CNNs) has expanded the capabilities of vision-based fire detection. Deep learning methods make fire detection

models possess the capability of learning complex and non-linear features from data, making them highly effective in fire scenarios. Flame images contain rich information, offering insights not only into fire position and area but also into the heat properties of the fire (Wang et al., 2023). Numerous CNN-based objective detection and classification models, such as Faster-RNN (Ren et al., 2017), VGG (Qian & Woodland, 2016), SSD (Liu et al., 2016) and YOLO series (Sohan et al., 2024), are utilized and improved as proven effective in fire detection missions (Bu & Gharajeh, 2019). Among these, YOLO family stands out for its superior detection performance and accuracy on flame detection with strong generalization, thus is extensively used (Casas et al., 2023).

After knowing the targets of firefighting missions, firefighting robots must figure out where they are and where to go, which is a state estimation and map-building task. Robots extract meaningful data by interpreting the data from their sensors and accessing the environmental conditions. While high-precision positioning can be achieved using Global Navigation Satellite System (GNSS) devices, this approach is impractical for indoor firefighting operations (Wang et al., 2020). Probabilistic map-based localization provides a method for estimating positions by calculating the probabilities of being at specific locations. Based on probabilistic conception, robots can localize themselves from an unknown starting position and track multiple possible positions (Panigrahi & Bisoy, 2022). However, this necessitates a pre-input environment map as a baseline. More recently, SLAM methods are widely researched, which make autonomous map building and robot position estimation possible. Many SLAM algorithms have been proposed for different sensors such as monocular cameras, binocular cameras, depth cameras and LiDAR. For SLAM algorithms, depth information is essential in estimating sensor pose and feature depth. LiDAR can easily acquire distance information and structural features, and its high reliability and invariance to illumination make it suitable for diversified conditions. However, the high expense and power consumption of LiDAR drive a preference for vision sensors in field applications (Pang et al., 2019). Monocular SLAM requires a procedure to create an initial map because depth cannot be recovered from a single image (Mur-Artal et al., 2015). It suffers from scale drift and may fail when being static or performing pure rotations in exploration (Cheng et al., 2022). Stereo cameras can solve these issues and simplify depth estimation. Despite these benefits, the expense of stereo cameras poses a budgetary constraint for engineering applications. Compared to the cheapest monocular camera, a stereo camera costs a few hundred dollars as well as LiDAR may reach several ten thousand dollars. Meanwhile, some shortcomings limit SLAM's application and popularization. These include challenges in initial pose estimation, increased computational cost, and susceptibility to feature mismatches or loss in complex environments (Guan et al., 2021).

2.2. Autonomous navigation of robots

To perform successful navigation, robots must pass through path planning and path tracking following the perception phase. The path planning phase lets robots calculate desired paths, and then robots achieve targets by modifying motor outputs in the path tracking phase (Ullah et al., 2024). Path planning can generally be categorized into global and local path planning according to the level of environmental information available. Global path planning assumes a static environment where robots have full knowledge of their surroundings. In contrast, local path planning is designed for robots that are in partially or completely unknown to the environment (Mohanty Prases K.and Singh, 2022). This allows robots to adapt to real-time changes by continuously monitoring their surroundings and responding accordingly. There is no fundamental distinction between global and local path planning since they can be applicable to each other, with the choice depending on the specific working environment (Liu et al., 2023).

Though path planning methods can be classified into graph search algorithms, bionic algorithms and AI algorithms, graph search algorithms are often favored in research and practical applications due to their stability and maneuverability (Zhou et al., 2022). Different graph search algorithms can be used to determine the lowest-cost path. This method determines a path from starting to target point by evaluating neighbor nodes within a grid map (Robotin et al., 2010). Dijkstra algorithm is a classic approach to search for a single-source optimal path based on traversal costs. Its blind search strategy often results in computational overhead and resource wastage (Debnath et al., 2019). A* search algorithm is an improved version of the Dijkstra algorithm by incorporating a heuristic function, which helps prioritize adjacent nodes for expansion. The heuristic function can largely reduce computational load and processing time by continuously calculating the least-cost path between the start node and the next node, preferring the path where the heuristic value is lower than the actual cost (Fransen & van Eekelen, 2023).

Path tracking is the last stage for robots to reach the targets, which requires smooth and accurate motion commands along the desired path. Proportional-integral-derivative (PID) controller is based on an error-feedback-based control loop mechanism and has been used for several decades in industrial applications for decades. Their wide-ranging popularity lies in the robustness, simplicity, good performance and ease of tuning control parameters (Shah & Agashe, 2016). Despite these advantages, the PID controller is also faced with challenges in tuning suitable controller gains for specific working conditions (Huang et al., 2020; Joseph et al., 2022). Geometric path tracking methods, such as the well-known pure pursuit method and Stanley method have been extensively developed and applied. The main advantage of these methods is the employment of a simple geometry model that generates timely feedback according to the robot's current states to meet the real-time requirement of autonomous navigation (Liu et al., 2021). The pure pursuit method, along with its variants, is among the most commonly used to solve path tracking problems in low-speed scenarios because of its resemblance to human driving behavior that maintains a vision down the road in advance for smooth driving (Kim et al., 2024).

2.3. Development of firefighting robots

Firefighting robots are specialized to serve in firefighting operations, carrying water jets and other firefighting equipment. In a broader sense, firefighting robots involve ground-based robots and unmanned aerial vehicles (UAVs) that perform firefighting operations. Robots have appeared in the firefighting field for over 30 years since the Tokyo Fire Department first used the robot named Rainbow 5 to assist with fire rescue in 1986 (Wang et al., 2024). This Rainbow 5 robot is based on a crawler chassis equipped with a two-nozzle water cannon, gas sensors and visual cameras. As shown in Fig. 1, the current commercial firefighting robots still depend on manual remote control, where operators are in the vicinity of robots. This is one of the most mature and common control strategies, but also limits the function of firefighting robots. The remote control operations entail visual guidance, which means the motion of robots is confined to the visual field of operators in most usage scenarios. When performing out-of-sight firefighting activities, the narrow view of embedded cameras is tough to respond to unexpected situations (Liu et al., 2016).

The progress of information technology and CV knowledge has contributed to the evolution of program-assisted control methods in firefighting robots. The program-assisted control allows robots to operate under a preprogrammed procedure, which is widely used in logistics and construction robots (Ye et al., 2024). Various sensors are installed to monitor temperature and gas concentration in environments, which can also be used as fire detectors. Some robotic products introduce the features of one-touch deployment and trajectory tracking function, helping with the building facility maintenance and inspection (Duan et al., 2023). These program-assisted firefighting robots achieve partial automation and suppress flames by adjusting pan-tilt-zoom (PTZ) apparatus without external manual commands (Zhao et al., 2024).

Fig. 1. Some applications of firefighting robots. (a) Guard Mark II, by Widemount Dynamics Tech Ltd. (b) Thermite RS3, by Howe & Howe Technologies (the image is licensed under a Creative Commons License, link: https://www.flickr.com/photos/lafd/53869591648/). (c) LUF 60, by LUF Ltd. (the image is licensed under a Creative Commons License, link: https://commons.wikimedia.org/wiki/File:Luf_60_feuerwehr_koeln_nebel.jpg). (d) SAFFiR, by Virginia Tech (the image is licensed under a Public Domain Dedication, link: https://nara.getarchive.net/media/john-seminatore-a-graduate-student-at-virginia-tech-92671c).

However, this strategy expects to program all activities, restricting robots to work in a specialized configuration instead of responding to dynamic environments (Chen et al., 2023). Catering to every possible change in the environment is an onerous and unsustainable approach. The program-assisted robot is limited to a static environment, such as warehouses and factories, with pre-defined environmental maps and parameters.

Autonomous firefighting robots are the most cutting-edge research direction in the field of firefighting, where worldwide researchers are committed to achieving fully robotic autonomy in fire detection, navigation and fire suppression. The primary task for autonomous firefighting robots is independent navigation, involving the abilities of flame positioning, robot localization, environmental map building, path planning, dynamic obstacle avoidance and path following. Given that multi-spectrum information can be obtained by mounting sensors on the robot, they directly return intuitive information about environmental information and confirm target position. Firefighting robots help firefighting commanders complete the analysis of fire environment deduction and prediction with the help of deep learning methods. When implementing fire extinguishing actions, robots autonomously achieve the target position and realize the autonomy of the robot operation with the aid of AI, servo control and other technologies.

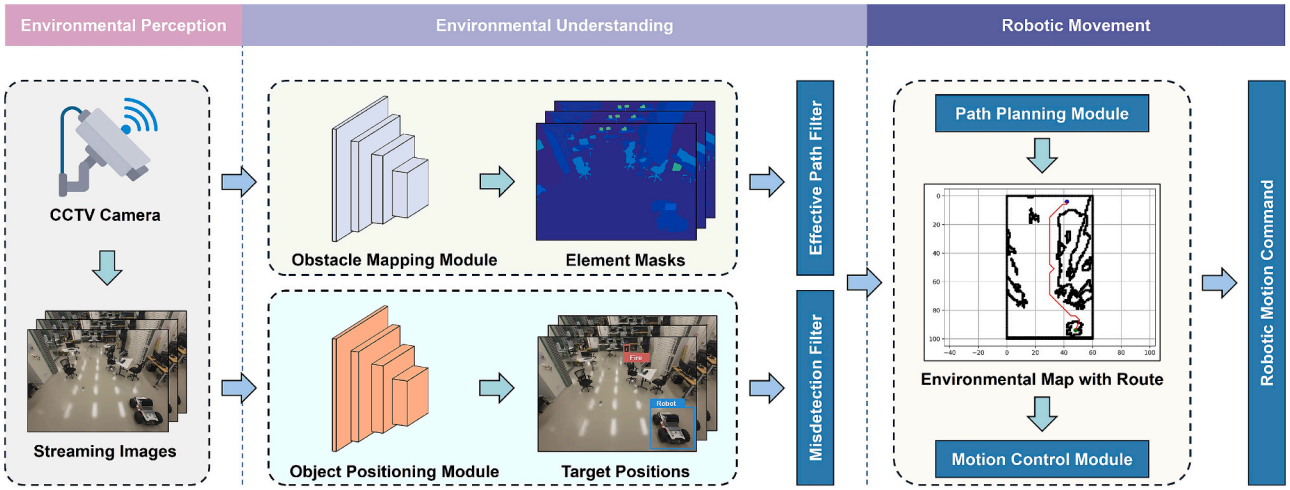

Fig. 2. Overall workflow of CCTV-informed autonomous robot navigation.

3. System design

3.1. Overview of the proposed framework

Orientation, destination and movement are the key factors of autonomous navigation of firefighting robots. Overall framework design is illustrated in Fig. 2. The execution of the whole system encompasses three phases: environmental perception, environmental understanding and robotic movement. This platform is implemented within the Robot Operating System (ROS) to facilitate the acquisition of robot states and the publication of movement commands.

The proposed framework contains four submodules: object positioning module, obstacle mapping module, path planning module and motion control module. The robot is completely imperceptive without any detectors or cameras. Hence the environmental information collected by the CCTV camera is used to navigate the movement and direction of the robot. YOLO model and OneFormer model (Jain et al., 2023) respectively serve as the object positioning module and obstacle mapping module to gather the spatial distribution of the robot, flame and obstacles. After the inverse perspective mapping for the filtered path mask and the positions of the fire source and robot, a 2D obstacle grid map is generated. Then the bidirectional A* path planning algorithm is employed in the path planning module which promptly gives the robot the shortest global path and addresses the dynamic changes in fire scenes. For accurate path following, a pure pursuit controller serves as the robot's motion control module. This end-to-end system processes the image sequence streamed from the CCTV camera as input, producing an environmental map with object coordinates as the output.

3.2. Object tracking and obstacle mapping

In order to simultaneously locate and track both the robot and fire source, the YOLOv8 detection model is adopted as the object positioning module. Compared to other models in the YOLO family, the YOLOv8 strikes a decent balance between performance and processing rate (Sohan et al., 2024). Its main backbone network is inherited from YOLOv5 with the modification on the cross-stage partial (CSP) layer. This modified CSP layer, also known as CSP bottleneck with two convolutions (termed C2f bottleneck), allows for improved execution speed without sacrificing performance. YOLOv8 has proven its state-of-the-art balance between speed and accuracy, which is critical to the immediacy of our work. A dataset was customized in this case which consists of smoke, flames, and robot poses to fine-tune YOLO model. This step can empower it to achieve robust and high-precision robot localization and fire detection in a complex building background.

The OneFormer model is a universe multi-task segmentation framework built on transformer architecture. The transformer architecture contains a novel self-attention design that can process various tokens in parallel as well as self-attention block effectively capture contextual dependencies across both spatial and temporal dimensions (Wang et al., 2024). These features are appropriate to dynamically evolving fire scenarios while having lower computational complexity and higher performance (Li et al., 2024). Based on these advantages, the OneFormer framework is employed as the obstacle mapping module to extract the accessible path mask.

The procedures of these two submodules are illustrated in Fig. 3, and a pseudocode is given in Algorithm 1. The perception part utilizes two submodules that process input images in parallel to guarantee

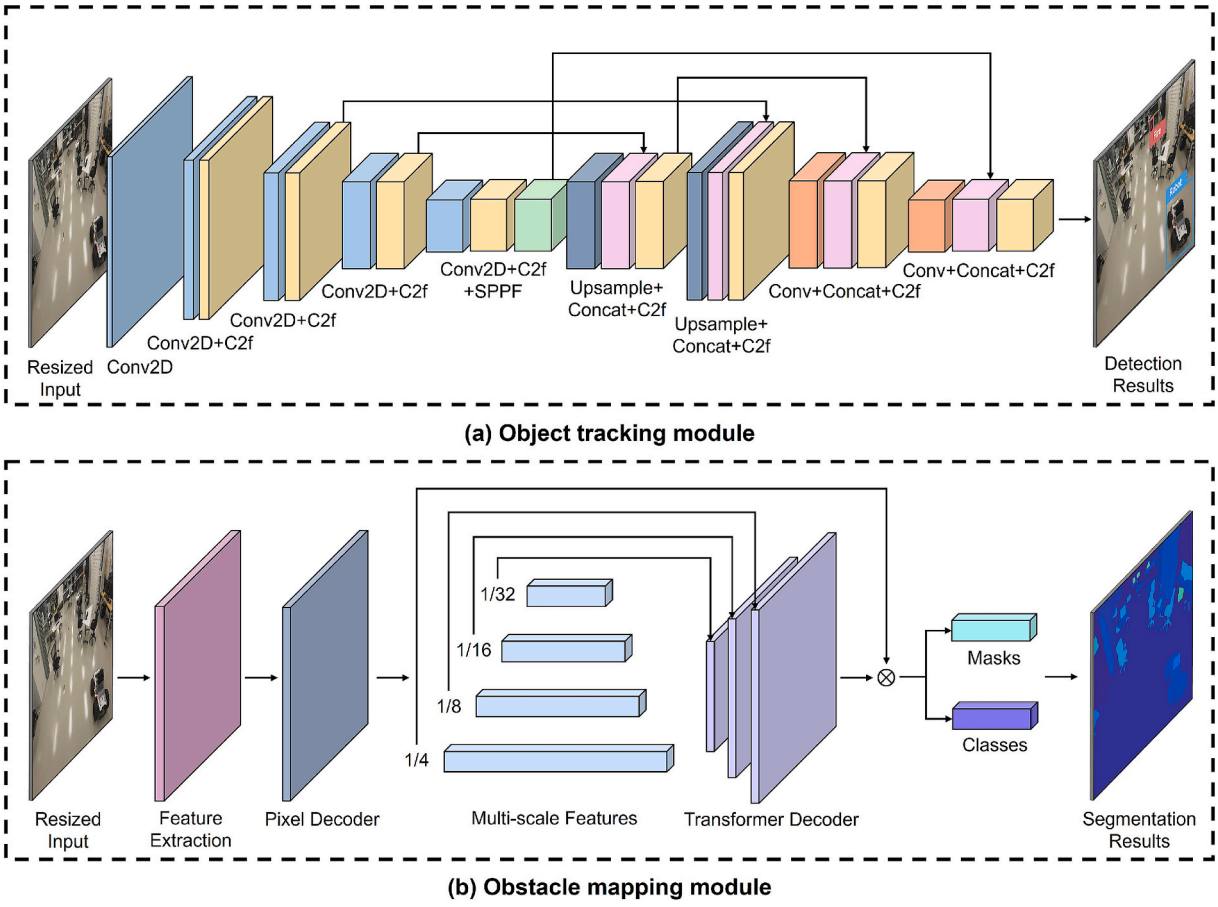

Fig. 3. Detailed architectures of object positioning module and obstacle mapping module. (a) Object positioning module modified from the YOLO structure. (b) Obstacle mapping module modified from the OneFormer structure.

synchronous processing and consistent output. Firstly, the object positioning module receives the video stream to filter the entity of interest, in this case is robot and fire source. The positions of these entities are then relayed to the path planning module. Concurrently, the obstacle mapping module segments the environment captured by the CCTV camera, classifying areas as either navigable terrain or obstacles. This process yields binary masks with precise boundary coordinates, which form a foundational environmental awareness. To create a structured and simplified representation for navigation, the perception results from both modules are integrated and rasterized into a 2D grid map that accounts for camera perspective.

Algorithm 1. Flame detection and environmental mapping

Input: *Streaming from CCTV camera* $I_C(x,y,t)$
Output: *Environmental map* $G(x,y,u)$ *with current positions of robot, flame and obstacle*
Initialize models $M_y = loader(w_y)$; $M_o = loader(w_o)$;
for *each image* $I(x,y)$ *in* $I_C(x,y,t)$ ***do***
Preprocess image $I_p(x,y) = processor(I(x,y))$;
Localize targets $L(b_i,c_i) = M_y(I_p(x,y))$;
Extract available path $P(m_i,c_i) = M_o(I_p(x,y))$;
end for
Filter error detections $L_a(b_i,c_i) = classifier(L(b_i,c_i))$;
Filter accessible path $P_a(m_i,c_i) = classifier(P(m_i,c_i))$;
Create 2D map $G(x,y,u) = mapper(L_a(b_i,c_i),P_a(m_i,c_i),M)$; *M is the perspective transformation matrix*;
return $G(x,y,u)$

Subsequently, since the CCTV camera footage is fixed and captured in a perspective view, it needs to be post-processed with an inverse perspective transformation to convert it into a 2D planar view. Given a point set $\{(x_1,y_1),(x_2,y_2),\cdots,(x_i,y_i),\}$ in the 2D grid and a set $\{(u_1,v_1),(u_2,v_2),\cdots,(u_i,v_i),\}$ *in* the real-world environmental map, the perspective transformation is expressed as

$$\begin{bmatrix} u_i \\ v_i \\ 1 \end{bmatrix} = M \begin{bmatrix} x_i \\ y_i \\ 1 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x_i \\ y_i \\ 1 \end{bmatrix} \tag{1}$$

where (u_i,v_i) is the homogeneous coordinate of target point in the environmental map, (x_i,y_i) is the coordinate of source point in the 2D grid, and w_i is the coefficient in homogeneous coordinates. The homography matrix M can be derived from

$$\begin{bmatrix} x_1 & y_1 & 1 & 0 & 0 & 0 & -u_1x_1 & -u_1y_1 & -u_1 \\ 0 & 0 & 0 & x_1 & y_1 & 1 & -v_1x_1 & -v_1y_1 & -v_1 \\ \vdots & & & & & & & & \\ x_i & y_i & 1 & 0 & 0 & 0 & -u_ix_i & -u_iy_i & -u_i \\ 0 & 0 & 0 & x_i & y_i & 1 & -v_ix_i & -v_iy_i & -v_i \end{bmatrix} \begin{bmatrix} h_{11} \\ h_{12} \\ h_{13} \\ h_{21} \\ h_{22} \\ h_{23} \\ h_{31} \\ h_{32} \\ h_{33} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 0 \end{bmatrix} \tag{2}$$

After applying this homography matrix, every point within the initial perspective grid, including entity masks, can be reprojected to the corresponding location on a new 2D environmental map. This transformation creates an orthographic view that is metrically scaled to real-world dimensions. Consequently, the robotic navigation task is significantly simplified. Path planning is no longer conducted in a complex 3D or perspective-distorted space but is instead reduced to a more tractable problem on a straightforward 2D rasterized map.

3.3. Path planning and motion control

For the path planning phase, local planning methods may be locally optimal, or even the target may be unreachable. Moreover, environment-aware sensors are one of the concerns of local path planning, which is not in conformity with the proposal of this paper. Therefore, the bidirectional A* algorithm, a global path planning method, is adopted to continually regenerate the optimum route during robotic movement. The bidirectional A* algorithm differs in that the node search simultaneously starts from both sides of initial and terminal points. It maintains separate open and closed lists of start point and target point, respectively denoting the node to be traversed and those already traversed. This strategy diminishes the sizes of the open and closed lists, thereby mitigating computational consumption (Xu et al., 2024). The comprehensive value $f(n_s,n_c,n_g)$ determines the priority of each node, which can be expressed as

$$f(n_s,n_c,n_g) = g(n_s,n_c) + h(n_c,n_g) \tag{3}$$

where $g(n)$ is the actual cost from the current node n_c to the start point n_s, and $h(n)$ is the heuristically expected cost from the current node n_c to the goal point n_g.

The node with the smallest $f(n)$ (means the highest priority) is selected from the priority queue as the next node to be traversed. The value of the traversal cost is determined by the Euclidean distance between the current and target nodes. Once the priority of all eight adjacent nodes near the current node has been computed, they enter the open list for the next iteration of lowest-cost selection. The algorithm inherently excludes points neighboring to obstacles and repeats the above procedure until it traverses to the current point as the end point. The A* algorithm was originally designed as a global planning approach, where the cost map is constant and not suitable for changing fire scenarios. Therefore, we modified the bidirectional A* algorithm by delivering a dynamic cost map that increases the passage cost of the corresponding blocked area in real time.

Configurations of PID control are featured by being specialized to a particular environment and are inappropriate to dynamic environmental changes. It may tend to overshoot or oscillate when navigating curved paths, especially under fire scenes. The pure pursuit controller realizes the awareness of geometrical look-ahead point which brings the abilities of smooth tracking for non-linear paths. This controller computes the required steering angle δ based on the geometric relationship between the robot's current position and a look-ahead point on the desired path. The steering angle is derived as

$$\delta = \arctan(\kappa L) \tag{4}$$

where L is the wheelbase of the robot. The value κ is the curvature between the robot and the look-ahead point, which is defined as

$$\kappa = \frac{2\sin\alpha}{L_d} \tag{5}$$

where α represents the heading error determined as the angle between the robot's current heading and the line connecting the robot to the look-ahead point, and L_d represents the look-ahead distance from the robot to the look-ahead point.

In the motion control module, the angular velocity is derived as the real-time non-linear adjustment based on the steering angle, while the linear velocity is constant to expedite target achievement. Since the perception submodules provide comprehensive environmental information, a curvature-adaptive control method is implemented to maintain motion stability and smoothness. The controller's primary feature is the dynamic adjustment of the look-ahead distance, which is linearly proportional to the calculated path curvature, as shown in the following equation

$$L_d = L_o + \frac{k_c}{C(s)+c} \tag{6}$$

where L_o is predetermined look-ahead distance, k_c is the weighted coefficient, and c is small constant to prevent the denominator from zero. This feature can avoid path deviation and excessive turning radius that may occur with fixed look-ahead distances on high-curvature segments. It enables the robot to dynamically adapt its cornering speed in response to different collision risk levels, enhancing navigational safety and precision.

Algorithm 2. Path planning and motion control

Input: Environmental map $G(x, y, c)$
Output: Movement commands with linear and angular velocities $C(v_{linear}, v_{angular})$

Receive current position from robotic odometry $Od = odometer(x, y, w)$; x and y are the odometry coordinates; w is the robotic yaw angle;
if reach_target is True **do**
$C(v_{linear}, v_{angular}) = (0, 0)$;
else
Initialize $R \leftarrow \{\}$; $Rt \leftarrow \{\}$; $Fl \leftarrow \{\}$; $Ob \leftarrow \{\}$;
for x_i, y_i, c_i in $G(x, y, c)$ **do**
if $c_i == 'fire'$ **do**
$Fl \leftarrow x_i, y_i$;
elif $c_i == 'robot'$ **do**
$Rt \leftarrow x_i, y_i$;
elif $c_i == 'path'$ **do**
$Ob \leftarrow x_i, y_i$;
end if
end for
Calculate desired path $R = bi_astar(Fl, Rt, Ob)$;
Pass path to controller $C(v_{linear}, v_{angular}) = pure_pursuit(R, Od)$;
end if
return $C(v_{linear}, v_{angular})$

Algorithm 2 outlines the detailed steps during the path planning and motion control processes. To ensure time-sensitive robot navigation and asynchronous processing, the motion control module is operating in parallel with the object positioning module, obstacle mapping module and path planning module using the multithread feature of Python. The motion controller updates movement commands once a new map and route are received from the path planning module. In view of the system operation latency, robotic odometry is utilized to estimate the robot's current coordinates, which are fed into the controller for real-time corrections. The route generated by the path planning module that is initially defined in the image coordinate of the CCTV camera is transformed into the odometry coordinate in the motion control module.

4. Experiments

4.1. Experimental setup

A DJI Action 4 camera with a suction cup mount is installed 4 m above the ground, serving as the CCTV camera for the experiments. Two non-perception robots of different sizes are used to test and validate the versatility and robustness of the proposed framework. One is a small educational robot equipped with multi-sensor devices (Fig. 4(a)), which are functionally disabled during the experiments. This robot is already fitted with a minicomputer for control. Another one is an AgileX Robotics SCOUT 2.0 robotic chassis that consists of motors, batteries and cases excluding sensors and control computers. Therefore, a laptop is integrated with the chassis using an aviation plug to USB port adapter (Fig. 4(b)).

Wireless Local Area Network (WLAN) is established to enable wireless intercommunication between robot, master computer and slave computer (Fig. 4(c)). As a part of building intelligent fire protection integration, the autonomous navigation system may have to serve multiple robots, hence is deployed on the master computer other than on a particular robot's edge device. The CCTV camera is in a wired connection to minimize communication delays. The Controller Area Network (CAN) bus is used to build communication between the robot and the slave computer. The slave computer receives and decodes the state messages of the robot and synchronously publishes motion commands through peer-to-peer CAN protocol. Because of safety policies at the experimental site, a campfire video is displayed on the monitor as a flame target.

Fig. 5 explicitly describes experimental environment settings and obstacle layouts. In none of these configurations does the robot have direct access to the fire source. For downscaled test settings, we set an area with a boundary of 2 × 5 m. Ethylene vinyl acetate (EVA) foams with a size of 20 × 20 cm are aligned as obstacles (see Fig. 5(a)). Fig. 5 (b) and Fig. 5(c) illustrate the experimental settings of real-scale scenarios, where the real-scale verification and dynamic responsiveness validation are sharing the same scene of 6 × 10 m. Several computer desks and office chairs are randomly placed in this scene. The projected size of computer desks is 1.5 × 0.5 m, and that of office chairs is 0.7 × 0.7 m. For dynamic changing environment, a new chair will be appended directly in front of the robot when moving for 6 s (see Fig. 5(c)).

4.2. Small-scale test experiment

Several 20 × 20 cm EVA cube foams are arranged as obstacles for the experiment. The linear velocity is constantly set to 0.3 m/s and angular velocity changes along with the calculated steering angle with a maximum of 1.0 rad/s. The upper limitations can avoid overshoot or undershoot behaviors. The look-ahead distance is fixed at 0.5 m for the motion controller to seek a suitable look-ahead point. The reason for using a scale model is that a narrow path poses a challenge to the motion control ability and system timeliness. All pre-existing sensors are disconnected from the robot to simulate the non-perception condition, retaining only its motion functions.

Fig. 6 shows the entire process of CCTV-informed autonomous navigation (see Video S1 for more details). The proposed platform is steadily tracking the robot and the flame while providing real-time feedback to adjust movement commands. At the beginning, the robot is unable to directly see the flame, as it is blocked by the EVA cubes. Traditional visual and laser detectors may fail under such a limited view, leading to target loss. When the vision from the CCTV camera is shared with the robot, the system can globally perceive the environment and devise the best strategy for the robot to achieve the target. The robot dexterously avoids obstacles and adjusts the angular speed accordingly. The entrance of the fire scene and the passable path are timely identified by the navigation method and the robot accurately moves along the

Fig. 4. Experimental settings. (a) Reduced-size educational robot (266 × 206 × 193 mm). (b) Large-size SCOUT robotic chassis (930 × 699 × 349 mm). (c) WLAN topological structure.

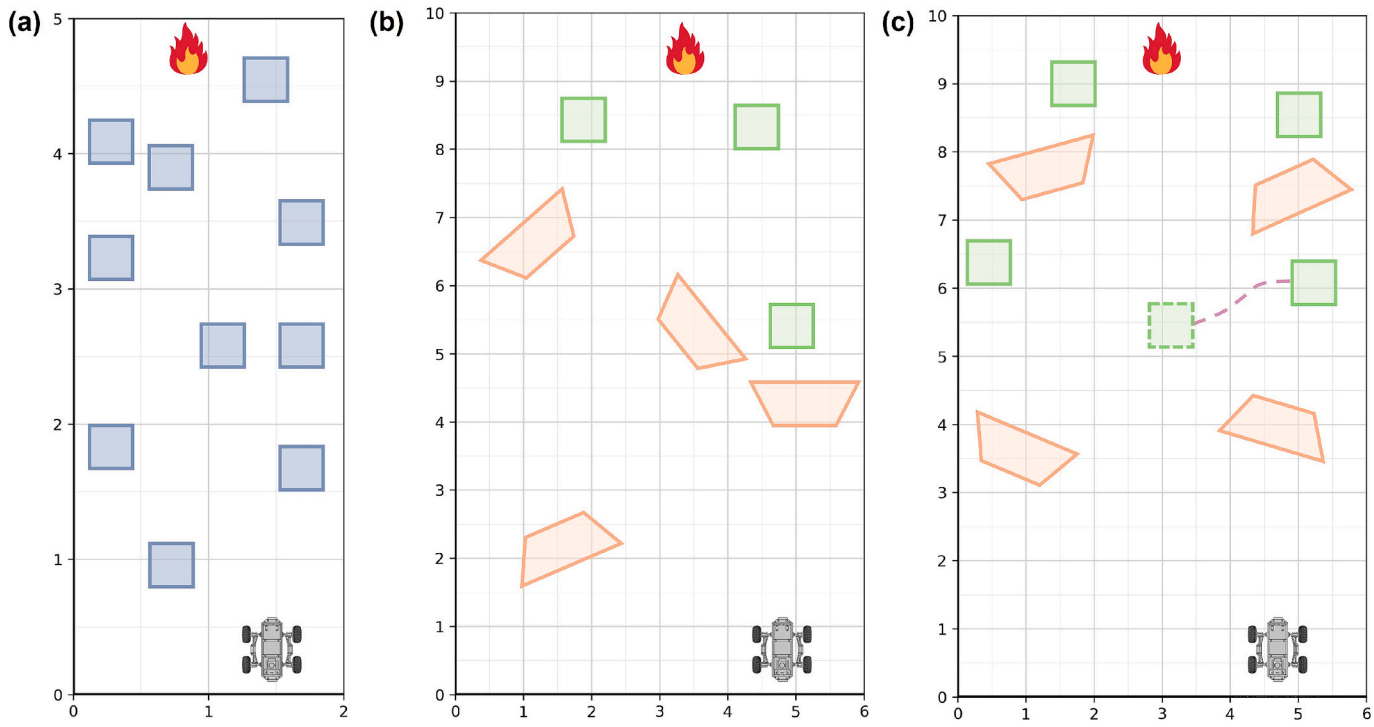

Fig. 5. Layouts of three experimental settings. (a) Downscaled test scene. (b) Real-scale validation scene. (c) Dynamically changing test scene, where purple dashed line is obstacle trajectory and dashed square is the position in the environment after addition.

calculated route. Despite operating in a confined space, the robot reaches the flame without colliding with the EVA cubes.

The environmental map synchronously generated by the framework is shown in Fig. 7, which illustrates the detailed trajectory of the robot and the distribution of obstacles. By defining the dimensions of the grid map, the proposed strategy enhances accuracy of robot localization and movement. The map is set to 25 × 55 corresponding to the real scale of fire scene where one grid means 1 × 1 cm. After 2D reprojection for the fire scene captured by the camera, the environmental map is coupled with the coordinate information of the robot, flame, obstacles and route. The path planning module constantly computes the desired route while the environmental map is updated by the object positioning module and obstacle mapping module. The incipient route follows an S-shape path and subsequent calculations result in basically similar route shapes. This indicates that the system can find the most accessible route at the initial stage and navigate the robot to the target position in the shortest possible time. This small-scale test demonstrates that the proposed system equips the robot with complete perception and understanding of the environment, enabling autonomous navigation in the constrained space. Even under low-speed working conditions, the system maintains high performance in robotic motion control and obstacle avoidance.

4.3. Real-scale validation experiment

A real-scale office layout is used to validate the generalization and practicability of the system. Some computer desks and chairs are placed randomly as obstacles within the fire scene. The size of real-scale field is 6 × 10 m where a grid size is 1 × 1 cm in the grid map. Due to the high maneuverability and performance of the chassis, the linear velocity is 0.5 m/s and the angular velocity is limited to 2 rad/s, both higher than the corresponding values for the educational robot. Considering the cumbersome body and large turning radius of the chassis, the look-ahead distance is set to 1 m. A longer look-ahead point affords the chassis the opportunity to anticipate and execute early adjustments at higher speeds. The chassis is capable of immediately fine-tuning its motion trajectory when encountering sharp or consecutive bends.

Fig. 8 shows that the flame is concealed by the obstacles while the chassis lacks environmental perception devices (Video S2 presents more details). The common navigation methods necessitate early exploration of the fire environment and localization of the fire source under such an unknown setting. However, with the CCTV camera providing a dominant view, the approach allows the chassis to acquire the necessary environmental information without performing pre-mapping activities. The system initialization costs approximately 3 s to load modules, activate communication and calculate the initial route. The chassis takes corresponding behaviors of rapid movement and precise obstacle avoidance under the guidance from the CCTV camera.

The environmental map in Fig. 9 provides an interpretable view of the CCTV-informed autonomous navigation framework in action. After 3 s of the system initialization, the chassis starts to move forward to the target flame following the desired route. The robotic chassis approaches the desk at 8 s and needs to immediately implement corresponding actions to avoid the collision. At 10 s of the experiment, the chassis takes a large steering angle and circumvents the malfunction caused by the collision risk. At 18 s, the flame lays on the right of the chassis, requiring another turn. As anticipated by the platform, the chassis moves along the edge of the desk and successfully manages to pass through the obstacle and reach the flame by 20 s.

4.4. Dynamic responsiveness validation

During a firefighting operation process, the changes and movements of obstacles can jam the trafficable path and impede the passage of firefighting equipment. Therefore, dynamic response capability is crucial for the autonomous navigation platform in fire scenarios. In this dynamic responsiveness validation experiment, the experimental settings are similar to the real-scale validation experiment, with the linear velocity of 0.5 m/s, the angular velocity of 2 rad/s, the look-ahead distance of 1 m and the experimental field size of 6 × 10 m. At the 6th second, a new chair will be added to the scene at the speed of 1 m/s, which is a relatively fast dynamic challenge for navigational adaptability and flexibility.

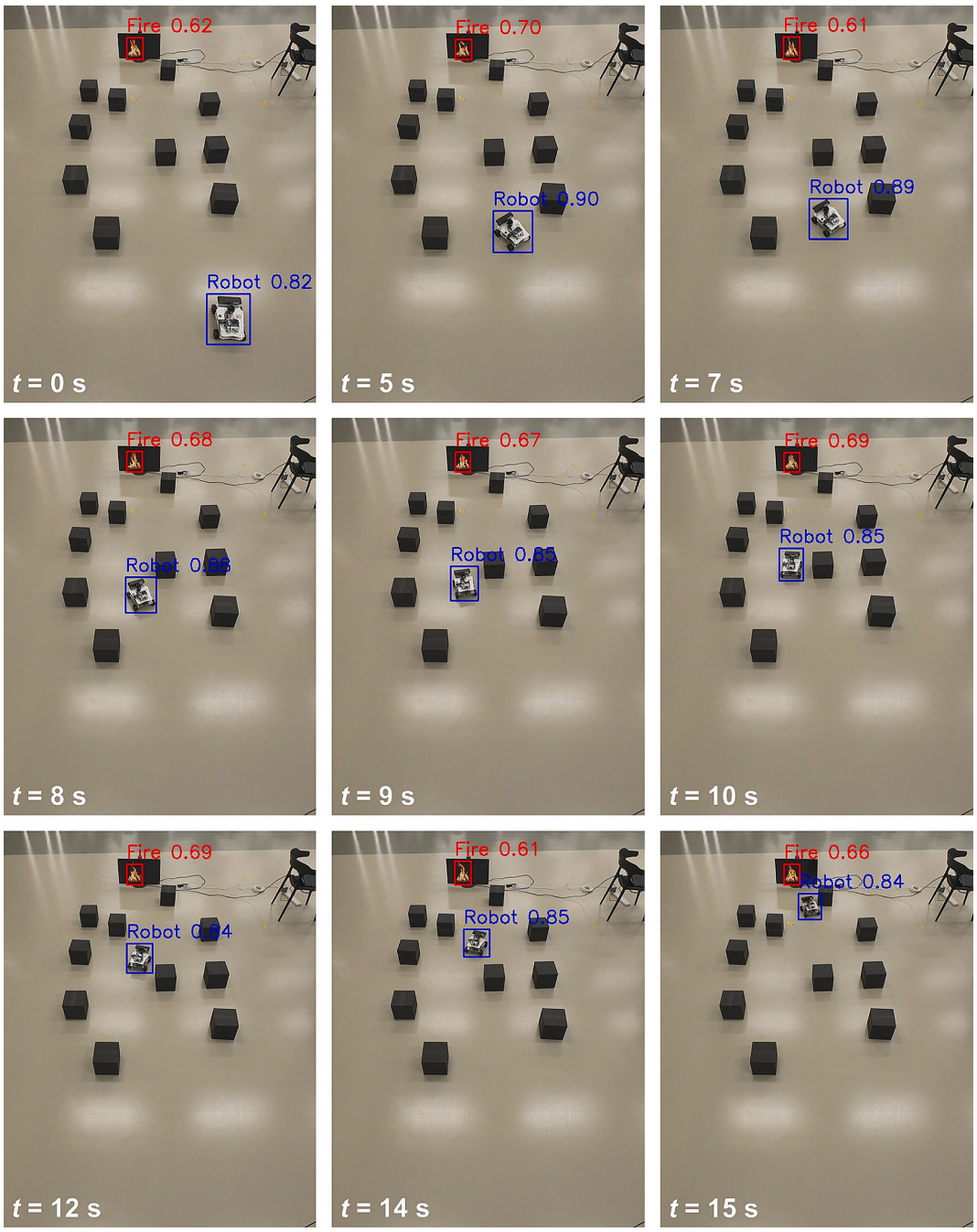

Fig. 6. Realistic scene of autonomous navigation in downscaled scenario (see Video S1 for more details).

As shown in Fig. 10, the approach spends 1 s for activation and initialization before the chassis begins moving (see a demonstration in Video S3). As the robotic chassis travels for 6 s, a chair is added to the fire scene, obstructing the robot's path. Upon nearing the chair at 7 s, the navigation system instructs the chassis to make a large turn to avoid this new barrier at 9 s. At 10 s of the travel, the chassis has successfully evaded the chair and reoriented itself towards the flame. Then the chassis moves along a new route eventually, ultimately reaching its goal despite the dynamic changes in the environment.

The 60 × 100 environmental map presented in Fig. 11 visualizes a more intuitive visualization of the entire navigation process. At the beginning of the navigation, the system calculates a similar route within the first 5 s, which indicates this route is the globally optimal solution based on the current environment. By 6 s of the navigation, the framework detects the newly introduced chair and updates the environmental map accordingly. A new route is then obtained according to the updated environmental information, including a left turn in front of the new chair. The robotic chassis successfully maneuvers around the new obstacle at 10 s. This result demonstrates that the navigation system is provided with a fast response ability to the dynamic fire environment. With the use of real-time observation of the CCTV camera and synchronous updates of environmental information, the robot can perform adaptive motions to circumvent dynamically movable obstacles within the environment.

4.5. *Conceptual and performance analysis*

In order to explicate the benefits of our CCTV-informed navigation strategy, this subsection undertakes a conceptual and baseline comparison in terms of cost efficiency, sensor independence and system

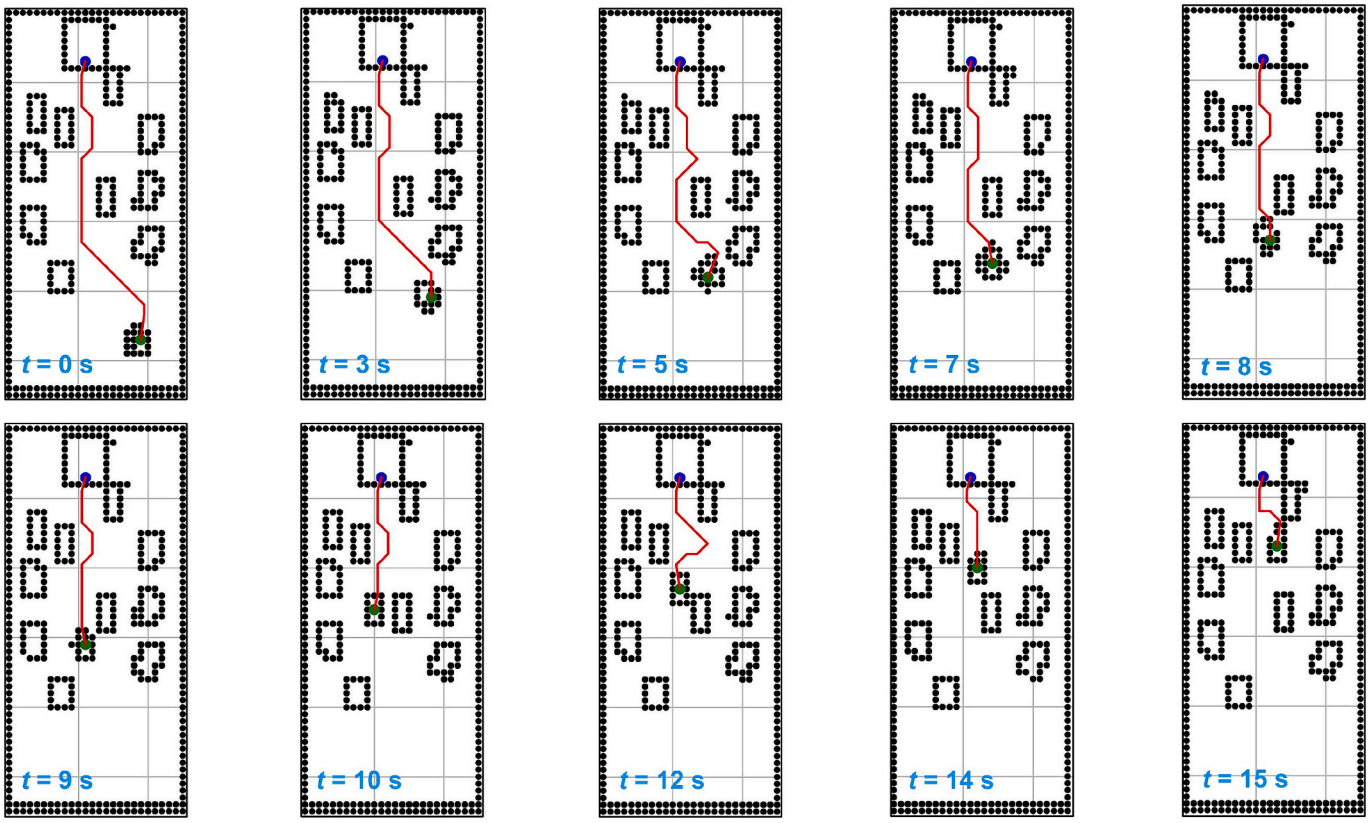

Fig. 7. Environmental map labeled with the coordinates of robot, flame and route. Green point represents the robot position (start node), blue point represents the flame position (target node), and red line represents the desired route.

accuracy. As described in Table 1, the CCTV-based guidance proves a significant cost advantage. Unlike SLAM methods that require precise and expensive instruments, our method exclusively leverages surveillance cameras already installed within buildings for map creation and navigation, making it highly extensible and universal in any environment with CCTV coverage. In addition, the centralized processing pattern of CCTV-based navigation eliminates the need for powerful and energy-intensive on-board computing units, such as DJI Manifold, Raspberry Pi and Jetson Nano. This substantially reduces power consumption. In contrast to current vision-based or laser-based SLAM methods, on-board sensors and computing units need to synchronously conduct environmental perception, spatial data alignment and map merging. This entails a high computational burden and electrical usage, which our CCTV-based strategy effectively offloads and circumvents.

Table 2 outlines the fundamental differences between the proposed CCTV-based navigation and SLAM workflows, particularly regarding system concept and sensor independence. A standard SLAM-guided robot functions as a self-contained entity that integrates perception, decision-making and execution. Its success is therefore critically dependent on the continuous and accurate operation of its local sensors, making sensor failure a single point of robotic blindness or even paralysis.

Our methodology overcomes this vulnerability by decoupling robotic motion capability from its on-board sensing capability. In this paradigm, robots are simplified actuators, responsible only for receiving instructions from the host server and performing movements. The complex task of perception and planning is offloaded to a central brain (i.e. a host server that processes a global view from the entire CCYV network). This centralized architecture yields two significant advantages. First, it offers a global and allocentric perspective of the operational environment. The system maintains simultaneous visibility of all robots, their positions, their states, and macro-environmental changes. This holistic awareness is crucial to achieving globally optimized multi-robot coordination, including dynamic path planning, task assignment and traffic management. While as for SLAM robots, their local egocentric view is limited to immediate vicinity and is susceptible to blind spots. Second, this approach inherently augments system redundancy, since it can instantly switch to an alternative camera view or auxiliary awareness devices to maintain navigational functions when sensor failure. It is a characteristic not available to an isolated SLAM-based unit.

Since the CCTV network provides global and high-frequency localization information, a common comparison of robotic trajectory and ground truth trajectory would not be suitable for performance evaluation. Typically, navigation methods conduct evaluation by comparing robotic trajectory with a fixed ground truth, which is generally yielded by inertial navigation system (INS) or real-time kinematic (RTK) GPS data. However, in our architecture, the CCTV camera takes the role of indoor GPS as it constantly yields this ground truth localization and feeds it directly into the control loop for error correction. Therefore, we performed a deep quantitative analysis of the internal data to evaluate performance, focusing on metrics such as the dynamic tracking error and the efficiency of the path correction maneuvers.

We executed 20 repeated experiments to statistically assess our navigation design across four metrics, with results presented in Table 3. The system achieved a 95 % success rate with 19 successes of 20 trials, where the single failure was attributed to computational congestion that prevented the motion controller from adjusting instructions in time. This issue can be addressed in future work by implementing a dynamic resource reallocation mechanism. In terms of path planning latency, we found that our solution approximately performed over 200 recalculations over a 20-second navigation period, yielding a considerable average computation time of 9.85 ms. Regarding trajectory tracking accuracy, the optimal path generated in the preceding time step served as the reference to estimate the average lateral error, which determined

Fig. 8. Realistic scene of autonomous navigation in real scenario (see Video S2 for more details).

how correctly the robot follows the optimal path and deviation extent. For instance, Holmberg et al. (2022) achieved the accuracy of about 4.1 cm using LiDAR/SLAM in a similar indoor environment. Our navigation scheme demonstrated high following fidelity, with an average lateral deviation of 9.196 cm from the dynamically planned optimal path, which we believe is a competitive result considering our low hardware cost and extensive potential. Finally, the path efficiency, measured by the path length ratio, was 1.031. A value this close to 1.0 indicates that the robot followed the planned trajectory with minimal redundant movement, confirming the controller's effectiveness.

4.6. Perspective applications

The proposed autonomous navigation pipeline for firefighting robots offers a novel integration with building maintenance system and digital twin model (see Fig. 12). In recent years, smart building schemes have become more concerned by worldwide governments. The digital twin enables to present building layout in a 3D form and be conveniently integrated with safety management to present more fire-related information (Chen et al., 2021; Kim et al., 2024). As a component of building smart firefighting system, this autonomous navigation framework is deployed on the central server without the need of robot-end computational resource. Ubiquitous CCTV networks furnish low-cost and all-day visualization for building health and emergency management.

Through Internet of Things (IoT) and 5G networks for data communication, the building digital twin program can intuitively illustrate the position of fire risks in the 3D building model and schedule patrol robots for proactive firefighting measures. Autonomous robot navigation with multi-camera inter-coordination can create a robotic cluster to facilitate facility inspection and emergency response. With the help of AI technologies, the heat release rate (HRR) and risk level are synchronously estimated, assisting occupant evacuation and dynamic analysis. Based on these metrics, fire commanders make more pragmatic judgements and tailored decisions to avoid possible casualties. Meanwhile, the water jet connectors reserved on patrol robots are compatible with firefighting trucks, reinforcing firefighting efficiency and shortening dispatch time.

The proposed autonomous navigation solution provides a promising paradigm for multi-robot firefighting coordination. By leveraging cross-region CCTV networks, the system gains a commanding, building-wide perspective with access to synchronously updated video feeds from diverse locations. This allows a centralized server to simultaneously detect and track the attributes and states of all key entities, such as robots, pedestrians, and fire hazards, and issue appropriate commands to the robotic cluster. This centralized, CCTV-guided approach fundamentally simplifies multi-robot operations. Unlike distributed systems that require each robot to build its own local map and engage in complex map-merging and state-sharing algorithms, this architecture provides a unified global environmental map. The central planner can construct this map through cross-camera alignment, track all robot trajectories in real-time, and efficiently compute collision-free paths for the entire cluster. Consequently, this information-centric model is ideal for optimized task allocation and traffic management. The central controller can facilitate functional synergies. For instance, by dispatching the nearest robot to a new point of interest, assigning distinct exploration zones to maximize coverage, or coordinating specialized units, such as a fire

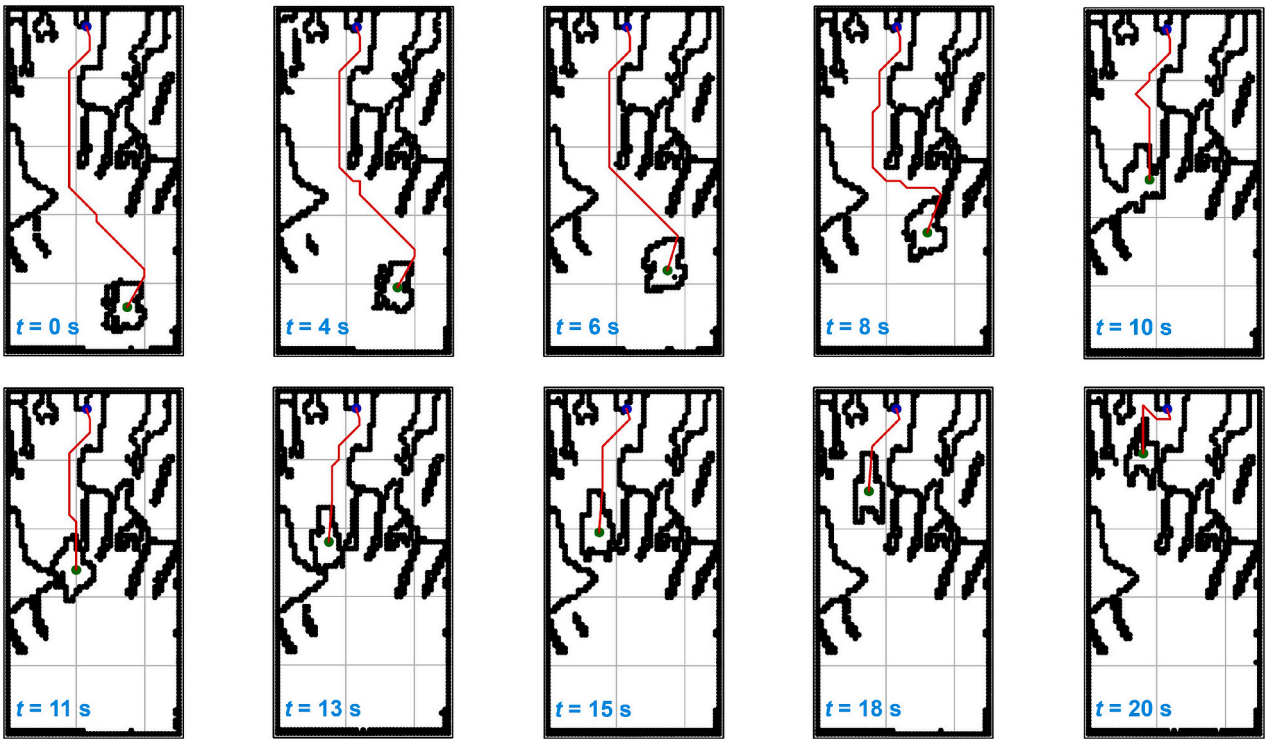

Fig. 9. Environmental map for autonomous navigation in a real scenario. Green point represents the robot position (start node), blue point represents the flame position (target node), and red line represents the desired route.

suppression robot and a support robot, to work in tandem.

As presented in Fig. 13, this proposed autonomous navigation scheme also provides a technological paradigm for smart firefighting protocols. Traditional fire response procedures rely on preliminary assessments conducted by firefighters upon arrival. This workflow is often hampered by the limited and subjective information provided by alarm callers, who may not be able to accurately describe the fire's extent. Furthermore, the response time of the fire brigade is subject to traffic conditions, introducing potential delays. Through the integration of the automated robotic protocol, the robot located inside the building can be immediately guided to the alarm point by CCTV cameras. It allows to transmit real-time video and critical on-site fire data (such as temperature, HRR and pedestrian distribution) back to the command center. Equipped with a portable fire extinguisher, the robot can perform autonomous initial suppression, effectively mitigating the fire in its incipient stage. For large fire scenarios that exceed the robot's standalone capabilities, its reserved water cannon interface is compatible with standard firefighting equipment, thereby augmenting the overall efficiency of the firefighting operation.

4.7. Challenges and future work

The primary challenge in enhancing the autonomous navigation pipeline is operation delay, as incorporating deep learning architectures takes substantial processing resources and runtime. Navigation tasks raise an urgent necessitation for prompt information transmission. However, when robots are in motion, data blockage occurs as the YOLO-based detector and Transformer-based architecture process tardive target coordinates. This lag consequently impacts the computation of the desired path. For experimental robot velocities, the system latency in this work is tolerable at about 0.5 s. However, this procrastination potentially leads to the maladjustment of firefighting robots for drastically changing settings in extreme fire circumstances. According to our tests, the instant update of Transformer-based element segmentation occupies approximately 70 % computational budget. Additionally, the CCTV camera video is streamed to the host computer in raw format, which may result in data congestion. It can be trusted that more lightweight algorithms and more effective data pre-processing approaches can further alleviate operation latency and bring more timely navigation guidelines without sacrificing performance, which is the direction of our future work.

Another challenge is the invisibility caused by smoke in fire scenarios. CCTV cameras are visual cameras that receive the light reflected from the surface of objects. The smoky and heat situations of extreme fire sites may wreck the functionality of building facilities and interrupt the vision of visual sensors, intercepting field monitoring and data transmission of CCTV cameras. Although our framework is specifically designed for autonomous response to early fire source, the adaptability and reliability of the proposed navigation system under more complex fire conditions need additional validation and improvement. Several potential solutions may address this challenge, with thermal infrared cameras emerging as a cost-effective alternative. Existing literature demonstrates the successful application of building thermal surveillance for identifying individuals, hazards and inventory (Warsono Ibrahim, 2016; Wong et al., 2009). The long-wave infrared (LWIR) spectrum is particularly relevant due to its superior smoke penetration capability. Even in conditions of zero visibility, LWIR imaging can effectively resolve the silhouettes of robots, obstacles, and heat-emitting fire sources, thereby enabling navigation where visual systems fail (Bianco et al., 2014; Owrutsky et al., 2006).

The insufficiency of this study is that the real-world validation was conducted in a single office environment. Although our experimental results suggest broader applicability, its performance in other types of real-world settings, such as those with complex floorplans that are hardly covered by a single camera, remains to be verified. These environments necessitate multicamera cooperative dispatch and robotic cluster coordination. Future work will aim to improve our system by developing cross-region re-identification and cluster control, and determine the framework's scalability and adaptiveness in more diverse and challenging environments, such as warehouses and factories. We

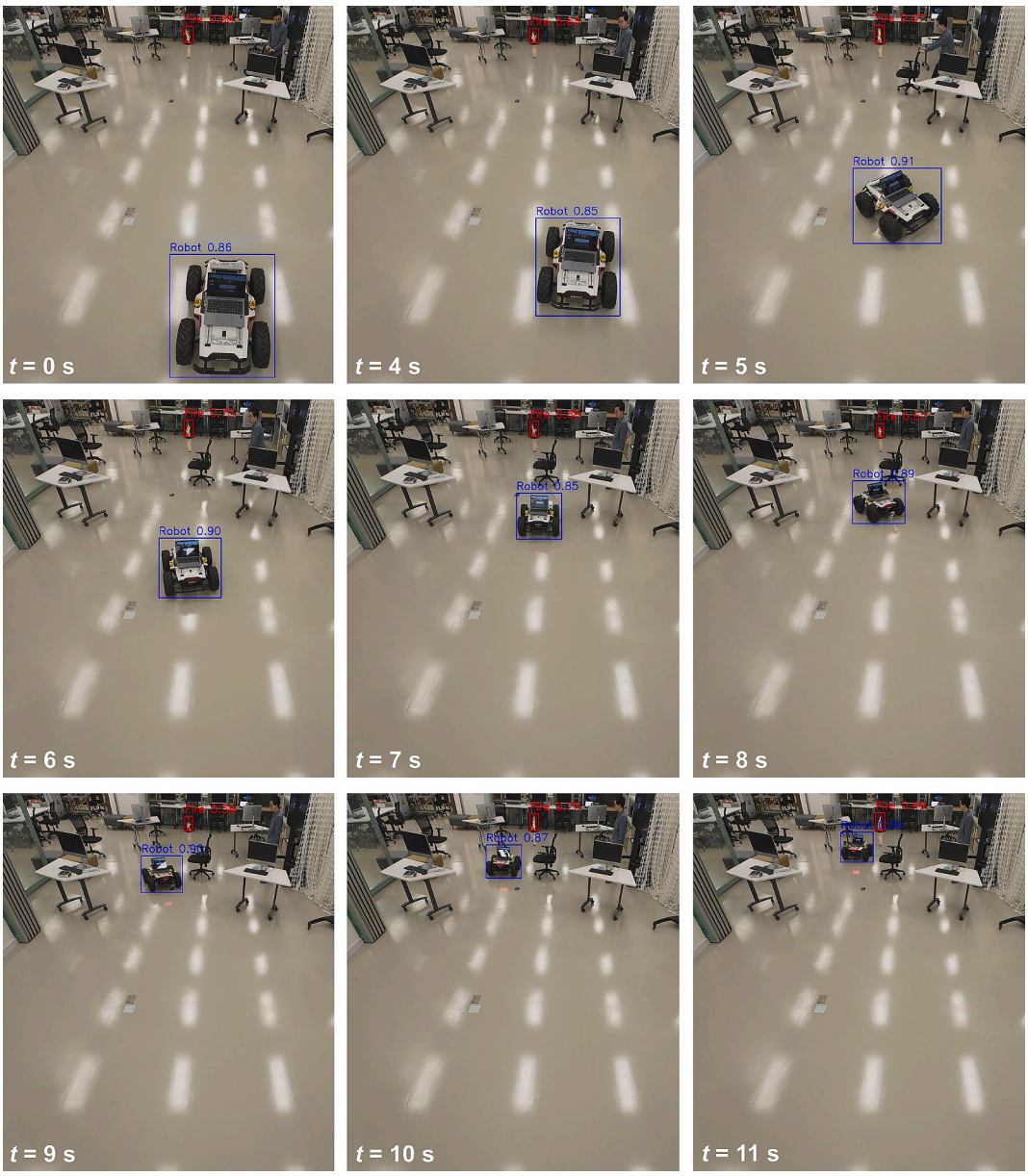

Fig. 10. Realistic scene of dynamic response ability validation (see Video S3 for more details).

also plan to investigate transfer learning techniques to allow the system to quickly adapt to new environments with minimal adjustment.

Furthermore, the footage delivered by previously installed CCTV cameras is blurry due to a combination of budget and maintenance considerations. When robots or flames are at a distance within camera views, the target positioning submodule might lose track since targets merely occupy a few pixels in low-resolution video streams. The emergence of small object detection and image super-resolution presents an innovative manner to augment the stability of the tracking function without replacing existing CCTV hardware, which can be investigated in the future.

5. Conclusions

This work proposes a CCTV-informed autonomous navigation design for firefighting robots. Through sharing vision between CCTV cameras and robots, this method can allow non-perception robots to receive and address environmental information in dynamically changing fire scenarios. This autonomous navigation framework overcomes the disadvantages of previous mapping and navigation methods that require preliminary input and exploration of environments. The dominant perspective of CCTV cameras can provide a simultaneously updating environmental map with visualized obstacle distribution and target location for firefighting robots. The performance and availability of the proposed framework are tested and validated under both low and high velocity operating conditions. Experimental results show that the navigation strategy can empower robots that are not sensor-equipped with the ability to sense and understand fire environments and guide automatic navigation to fire sources without third-party intervention. Furthermore, using the continuously streaming images from CCTV camera, the system constantly updates the environmental data and optimizes the desired route. Then firefighting robots can perform corresponding actions on the changes and additions of obstacles. In the validation experiment, the robotic chassis successfully avoids the new barrier and redirects to the target, proving the ability of the system to respond dynamically to fire scenes.

Due to the extensive presence of CCTV cameras inside each building, the proposed methodology enables a low-cost solution for firefighting

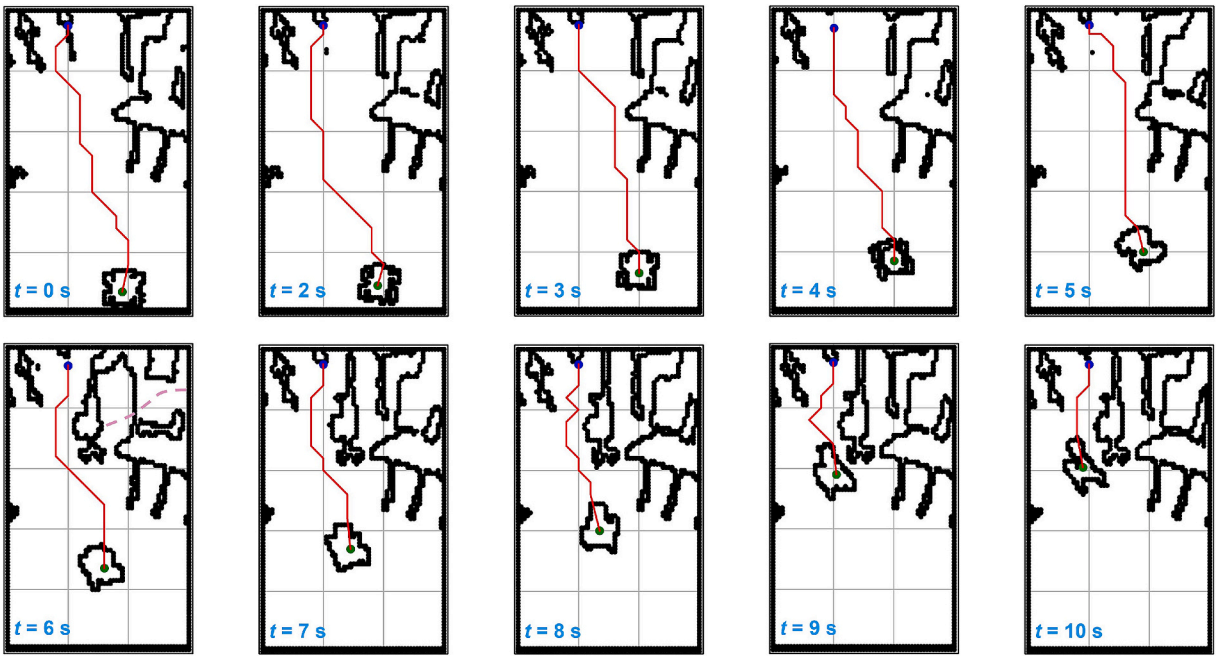

Fig. 11. Environmental map of autonomous navigation in a dynamic scenario. Green point represents the robot position (start node), blue point represents the flame position (target node), red line represents the desired route and purple dashed line represents the trajectory of dynamic obstacle.

Table 1
Comparison between the proposed CCTV-guided navigation and SLAM methods on cost efficiency.

Solution	Perception	Sensor expense	Computing unit	Power consumption
CCTV-based navigation	CCTV camera	~$ 0 (already installed in buildings)	Host computer	Low (only support robot movement)
Vision-based SLAM	Depth camera	$ 300 (Intel D435) $ 400 (ORBBEC Astra 2)	Local high-performance computer	High (support both robot movement and on-board sensors)
Laser-based SLAM	2D LiDAR	$ 460 (RPLIDAR S3) $ 400 (YDLIDAR TG50)		
	3D LiDAR	$ 2500 (Velodyne VLP16) $ 1400 (Livox Avia)		

Table 2
Comparison between the proposed CCTV-guided navigation and SLAM methods on senor independence.

Solution	Concept	Viewpoint	Independence	Redundancy
CCTV-based navigation	Central brain and simple actuator	Global optimal view	Motion capability decoupled from on-board sensing capability	Degrade gracefully
Vision-based SLAM Laser-based SLAM	Fully autonomous intelligent individual	Local egocentric view	Motion capability coupled with on-board sensing capability	Single point of failure

Table 3
Key metrics to evaluate the efficiency and accuracy of the CCTV-informed navigation.

Success rate	Average planning time (ms)	Average lateral error (cm)	Path length ratio
95 % (19/20)	9.85	9.196	1.031

patrol robots to automatically localize fire risks, move to targets and take related actions. Establishing vision sharing pipeline between CCTV cameras and firefighting robots significantly enhances the emergency response and reduces the demands of high-cost mapping sensors and computational resources. Firefighting robots can conduct autonomous navigation and realize proactive fire extinguishing as soon as headquarters confirm the occurrence of flames through CCTC cameras. They can effectively cooperate with current firefighting protocols and workflows, evolve fire suppression efforts into early fire intervention. In the future, this navigation system can be embedded into building digital twin models and collaborate with other building facilities, achieving automated full-cycle patrol inspection and fire suppression.

CRediT authorship contribution statement

Rong Deng: Methodology, Software, Investigation, Visualization, Writing – original draft. **Saizhe Ding:** Methodology, Investigation, Validation. **Yifei Ding:** Investigation, Validation. **Meng Wang:** Investigation, Validation, Resources. **Xinyan Huang:** Conceptualization, Methodology, Supervision, Writing – review & editing, Funding acquisition. **Asif Sohail Usmani:** Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial

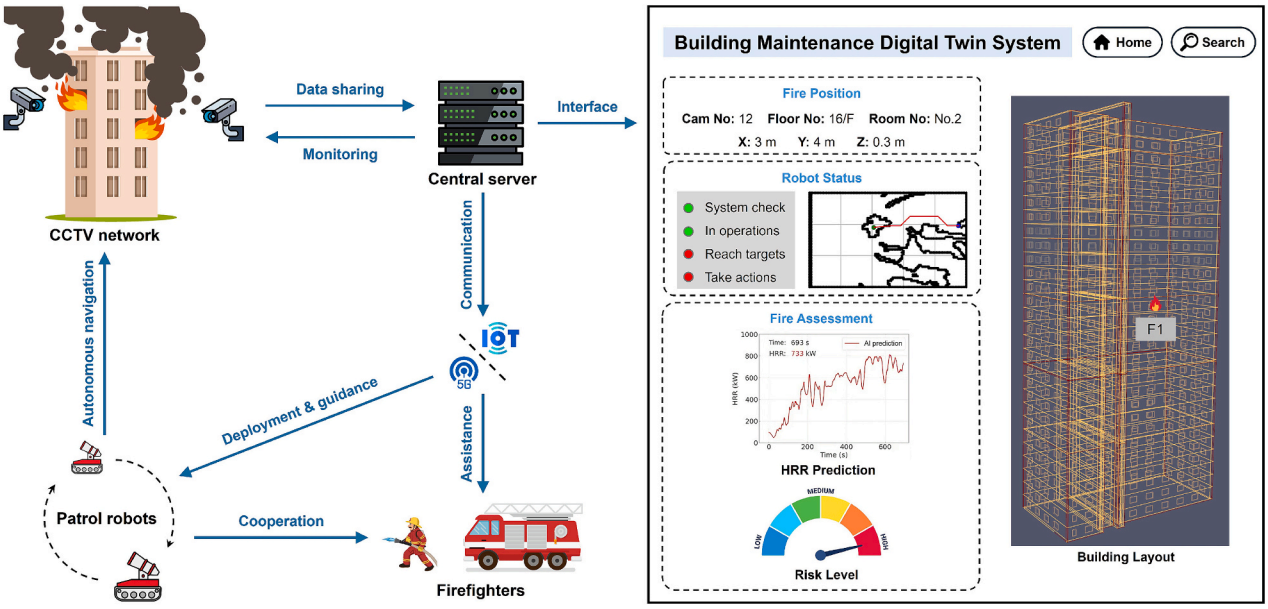

Fig. 12. Prospective integration with building maintenance and digital twin.

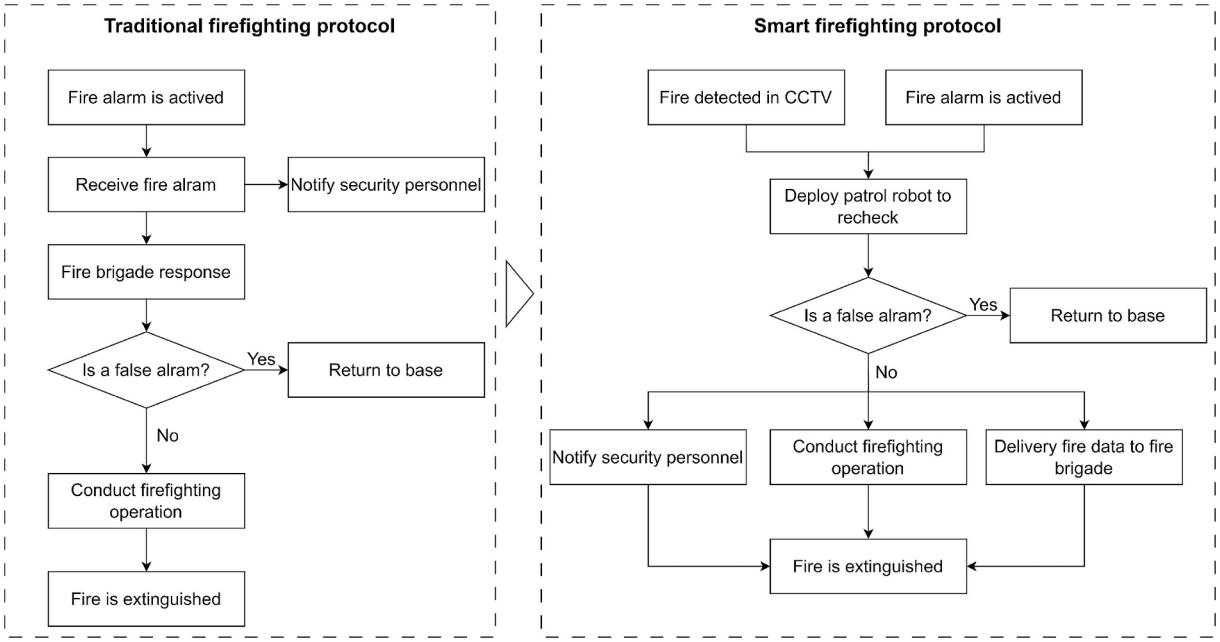

Fig. 13. Evolution of firefighting protocol towards smart workflow by being combined with the CCTV-informed navigation solution.

interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eswa.2025.129210.

Data availability

Data will be made available on request.

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